KAIZEN MATHS

Lesson starter

Differential Equations: First and Second Order

Modelling Problems

Name

S1

Form and solve each growth or decay model.

A population P changes at a rate proportional to P. P(0) = 300 and dP dt = 0.08P. Find P(t), then find P(12).

S2

Form and solve each Newton cooling model.

An object cools in a room at 18°C. Its temperature T satisfies dT dt = −0.08(T − 18) and T(0) = 80. Find T(t), then find T(10).

S3

Form and solve each mixing model.

A tank contains 120 litres of liquid. Brine containing 2 g/L of salt enters and leaves at 4 L/min. If S(0) = 40, form and solve the model for S grams of salt.

S4

Use the extra condition to find the model parameter.

A substance decays according to dM dt = −kM. M(0) = 100 and M(8) = 40. Find k and hence M(t).

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ALG-2BF2-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Differential Equations: First and Second Order

Modelling Problems

Name _____

EA1

Form and solve each growth or decay model.

A population P changes at a rate proportional to P. P(0) = 800 and dP dt = 0.03P. Find P(t), then find P(5).

EA2

Form and solve each Newton cooling model.

An object cools in a room at 18°C. Its temperature T satisfies dT dt = −0.08(T − 18) and T(0) = 100. Find T(t), then find T(20).

KAIZEN MATHS

Exit ticket

Differential Equations: First and Second Order

Modelling Problems

Name _____

EB1

Form and solve each mixing model.

A tank contains 80 litres of liquid. Brine containing 3 g/L of salt enters and leaves at 5 L/min. If S(0) = 40, form and solve the model for S grams of salt.

KAIZEN MATHS

Exit ticket

Differential Equations: First and Second Order

Modelling Problems

Name _____

EA1

Form and solve each growth or decay model.

A population P changes at a rate proportional to P. P(0) = 800 and dP dt = 0.03P. Find P(t), then find P(5).

EA2

Form and solve each Newton cooling model.

An object cools in a room at 18°C. Its temperature T satisfies dT dt = −0.08(T − 18) and T(0) = 100. Find T(t), then find T(20).

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KAIZEN MATHS

Exit ticket

Differential Equations: First and Second Order

Modelling Problems

Name _____

EB1

Form and solve each mixing model.

A tank contains 80 litres of liquid. Brine containing 3 g/L of salt enters and leaves at 5 L/min. If S(0) = 40, form and solve the model for S grams of salt.

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Teacher answers & working

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S1

Form and solve each growth or decay model.

A population P changes at a rate proportional to P. P(0) = 300 and dP dt = 0.08P. Find P(t), then find P(12).

Answer:

$P(t) = 300e^{0.08t}$, so $P(12) \approx 783.5$

Working:

Separate the variables.

$\frac{1}{P}\,dP = 0.08\,dt$

Integrate both sides.

$\ln P = 0.08t + C$

Write in exponential form.

$P = Ae^{0.08t}$

Use $P(0) = 300$.

$300 = Ae^{0}$

$A = 300$

Substitute t = 12.

$P(12) = 300e^{0.96} \approx 783.5$

$P(t) = 300e^{0.08t}$, so $P(12) \approx 783.5$

Teacher answers & working

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S2

Form and solve each Newton cooling model.

An object cools in a room at 18°C. Its temperature T satisfies dT dt = −0.08(T − 18) and T(0) = 80. Find T(t), then find T(10).

Answer:

$T(t) = 18 + 62e^{-0.08t}$, so $T(10) \approx 45.9°C$

Working:

Separate the variables.

$\frac{1}{T-18}\,dT = -0.08\,dt$

Integrate both sides.

$\ln|T - 18| = -0.08t + C$

Write in exponential form.

$T - 18 = Ae^{-0.08t}$

Use $T(0) = 80$.

$80 - 18 = Ae^{0}$

$A = 62$

$T(t) = 18 + 62e^{-0.08t}$

$T(10) = 18 + 62e^{-0.80} \approx 45.9°C$

$T(t) = 18 + 62e^{-0.08t}$, so $T(10) \approx 45.9°C$

Teacher answers & working

Differential Equations: First and Second Order · Modelling Problems

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S3

Form and solve each mixing model.

A tank contains 120 litres of liquid. Brine containing 2 g/L of salt enters and leaves at 4 L/min. If S(0) = 40, form and solve the model for S grams of salt.

Answer:

$S = 240 - 200e^{-\frac{1}{30}t}$

Working:

Rate in = 4 × 2 = 8 g/min.

Rate out = $4 \times \frac{S}{120} = \frac{4S}{120}$ g/min.

Model the rate of change.

$\frac{dS}{dt} = 8 - \frac{4S}{120}$

Rewrite in linear form.

$\frac{dS}{dt} + \frac{1}{30}S = 8$

The equilibrium value is 240, so the solution has the form $S = 240 + Ce^{-\frac{1}{30}t}$.

Use $S(0) = 40$.

$40 = 240 + C$

$C = -200$

$S = 240 - 200e^{-\frac{1}{30}t}$

Teacher answers & working

Differential Equations: First and Second Order · Modelling Problems

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S4

Use the extra condition to find the model parameter.

A substance decays according to dM dt = −kM. M(0) = 100 and M(8) = 40. Find k and hence M(t).

Answer:

$k = \frac{1}{8}\ln\frac{100}{40}$, so $M(t) = 100e^{-\frac{1}{8}\ln\frac{100}{40}t}$

Working:

Separate the variables.

$\frac{1}{M}\,dM = -k\,dt$

Integrate both sides.

$\ln M = -kt + C$

Write in exponential form.

$M = Ae^{-kt}$

Use $M(0) = 100$.

$M = 100e^{-kt}$

Use $M(8) = 40$.

$40 = 100e^{-8k}$

$\frac{40}{100} = e^{-8k}$

$\ln\frac{40}{100} = -8k$

$k = \frac{1}{8}\ln\frac{100}{40}$

$k = \frac{1}{8}\ln\frac{100}{40}$, so $M(t) = 100e^{-\frac{1}{8}\ln\frac{100}{40}t}$

Teacher answers & working

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EA1

Form and solve each growth or decay model.

A population P changes at a rate proportional to P. P(0) = 800 and dP dt = 0.03P. Find P(t), then find P(5).

Answer:

$P(t) = 800e^{0.03t}$, so $P(5) \approx 929.5$

Working:

Separate the variables.

$\frac{1}{P}\,dP = 0.03\,dt$

Integrate both sides.

$\ln P = 0.03t + C$

Write in exponential form.

$P = Ae^{0.03t}$

Use $P(0) = 800$.

$800 = Ae^{0}$

$A = 800$

Substitute t = 5.

$P(5) = 800e^{0.15} \approx 929.5$

$P(t) = 800e^{0.03t}$, so $P(5) \approx 929.5$

Teacher answers & working

Differential Equations: First and Second Order · Modelling Problems

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EA2

Form and solve each Newton cooling model.

An object cools in a room at 18°C. Its temperature T satisfies dT dt = −0.08(T − 18) and T(0) = 100. Find T(t), then find T(20).

Answer:

$T(t) = 18 + 82e^{-0.08t}$, so $T(20) \approx 34.6°C$

Working:

Separate the variables.

$\frac{1}{T-18}\,dT = -0.08\,dt$

Integrate both sides.

$\ln|T - 18| = -0.08t + C$

Write in exponential form.

$T - 18 = Ae^{-0.08t}$

Use $T(0) = 100$.

$100 - 18 = Ae^{0}$

$A = 82$

$T(t) = 18 + 82e^{-0.08t}$

$T(20) = 18 + 82e^{-1.60} \approx 34.6°C$

$T(t) = 18 + 82e^{-0.08t}$, so $T(20) \approx 34.6°C$

Teacher answers & working

Differential Equations: First and Second Order · Modelling Problems

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EB1

Form and solve each mixing model.

A tank contains 80 litres of liquid. Brine containing 3 g/L of salt enters and leaves at 5 L/min. If S(0) = 40, form and solve the model for S grams of salt.

Answer:

$S = 240 - 200e^{-\frac{1}{16}t}$

Working:

Rate in = 5 × 3 = 15 g/min.

Rate out = $5 \times \frac{S}{80} = \frac{5S}{80}$ g/min.

Model the rate of change.

$\frac{dS}{dt} = 15 - \frac{5S}{80}$

Rewrite in linear form.

$\frac{dS}{dt} + \frac{1}{16}S = 15$

The equilibrium value is 240, so the solution has the form $S = 240 + Ce^{-\frac{1}{16}t}$.

Use $S(0) = 40$.

$40 = 240 + C$

$C = -200$

$S = 240 - 200e^{-\frac{1}{16}t}$